Chapter 2

Dual space: bras, kets, adjoint operator

2.1 The Idea of Duality

Duality is the central thread running through this entire lesson. Let us give an intuitive account of the concept. In mathematics, duality encompasses many different situations, but in the context of Hilbert spaces it expresses a simple and profound idea: *a vector can be viewed not only as an object in its own right, but also as an operator acting on other vectors to produce numbers* (that is, as a linear form). In quantum mechanics, the number so produced is a probability amplitude.

The quantum formalism thus makes full use of these two facets, and *Dirac notation* provides an ingenious writing system for manipulating them with ease. It features *kets*, written $|u\rangle$, which represent vectors in the space, and *bras*, written $\langle u|$, which represent linear forms acting on those vectors. Their combination $\langle u|v\rangle$ produces a scalar called a *bracket* (or inner product), while expressions of the form $\sum a_{ij} |u_i\rangle \langle v_j|$ describe operators.

This notation has become the standard language of quantum physics, and mastering it fully is essential. A summary of the computational rules will be provided following this lesson. In practice, it is entirely possible to learn to manipulate these symbols without truly understanding the mathematics behind them. But to grasp their deeper meaning, one must undertake the programme we outline here: understanding how one constructs the complete identification between vectors and linear forms, and why this approach naturally leads to defining another essential object—the adjoint of an operator.

In the next section, we will attempt to establish an isomorphism between a vector space and its algebraic dual, defined as the set of *all* its linear forms. We will see that this is impossible. Indeed, although a finite-dimensional vector space is isomorphic to its dual, this isomorphism is not canonical (it depends on an arbitrary choice of basis). Moreover, the situation worsens in infinite dimensions: no such isomorphism exists at all. These obstacles invalidate the automatic identification between vectors and linear forms, and justify invoking the richer structure of Hilbert spaces.

Section 3 shows how to exploit this additional structure to overcome these difficulties. By using the inner product, and restricting to the continuous linear forms that constitute the so-called topological dual, one arrives at a remarkable result. The Riesz theorem asserts that for any Hilbert space, regardless of dimension, there exists a canonical antilinear isometric isomorphism between the space and its topological dual.

This result provides the rigorous foundation for Dirac notation, which we then detail in Sections 4 and 5 (kets, bras, and linear operators), before showing how the Riesz isomorphism naturally gives rise to the notion of the adjoint of an operator (Section 6). Section 7 summarises this construction in a comprehensive diagram. Throughout this lesson, we use the finite-dimensional case as a concrete example to illustrate the theory, working systematically over the field $\mathbb{C}$.

2.2 Linear Forms and the Algebraic Dual

Let E be a vector space over $\mathbb{C}$. Its *algebraic dual* E^* is defined as the set of **linear forms** on E, that is, linear maps $\varphi : E \to \mathbb{C}$.

In finite dimension n, fixing a basis $(e_1, \ldots, e_n)$ of E, one can construct a corresponding family of linear forms $(\theta_1, \ldots, \theta_n)$ in E^*, defined by the relation:

$$\theta_j(e_i) = \delta_{ij}$$

One then shows that this family is free and spanning, hence a basis of E^* (called the dual basis). It follows that $\dim(E^*) = \dim(E)$, which implies that E and E^* are isomorphic. One such isomorphism is the map T that associates to each vector $x = \sum x_i e_i$ the linear form $w_x = \sum x_i \theta_i$.

However, this isomorphism T is in a sense artificial: it depends on the initial choice of basis for E. Changing the basis changes the map T as well. There is therefore no canonical (natural) identification between a vector space and its algebraic dual.

In infinite dimensions, a classical result shows that E is never isomorphic to its algebraic dual, since the latter has strictly larger dimension (cf. the ErdHos–Kaplansky theorem). For instance, if E has countable algebraic dimension, its algebraic dual has uncountable algebraic dimension.

In both cases, there is no canonical identification between vectors and linear forms.

2.3 The Topological Dual and the Riesz Theorem

In a Hilbert space $\mathcal{H}$, the norm associated with the inner product allows one to single out a special class of linear forms: those that are *continuous*. One says that $\varphi : \mathcal{H} \to \mathbb{C}$ is continuous at x_0 if

$$\forall \varepsilon > 0,\ \exists \delta > 0 : \|x - x_0\|_{\mathcal{H}} < \delta \implies |\varphi(x) - \varphi(x_0)| < \varepsilon,$$

and if it is continuous at one point, linearity implies it is continuous everywhere[1]. The set of continuous linear forms constitutes a vector subspace of the algebraic dual. It is called the topological dual and is again (by an abuse of notation) denoted $\mathcal{H}^*$.

[1]These points will be revisited in subsequent lessons (topology, theory of linear operators)

Definition 2.1 : Topological dual

Let $\mathcal{H}$ be a Hilbert space over $\mathbb{C}$. The set of continuous linear forms on $\mathcal{H}$ is a vector space, called the *topological dual* of $\mathcal{H}$ and denoted $\mathcal{H}^*$. This space carries a natural norm, called the dual norm, given by

$$\|\varphi\|_{\mathcal{H}^*} \overset{\text{def}}{=} \sup_{\|x\|_{\mathcal{H}}=1} |\varphi(x)|,$$

with respect to which $\mathcal{H}^*$ is a complete normed vector space (i.e., a Banach space).

Remark: this norm is a special case of the operator norm encountered later (cf. the lesson on linear operators). The construction above is valid for any normed space. What makes the Hilbert case special is the following theorem: the inner product is used to construct a bijective identification between every continuous linear form and a vector.

Theorem 2.1 : Riesz Representation Theorem

Let $\mathcal{H}$ be a Hilbert space, separable or not. Every continuous linear form $\varphi \in \mathcal{H}^*$ can be written uniquely in the form

$$\varphi(x) = \langle u, x\rangle$$

for some vector $u \in \mathcal{H}$.

Note that the map $\Phi : u \mapsto \varphi_u = \langle u, \cdot\rangle$ from $\mathcal{H}$ to $\mathcal{H}^*$ is injective even in a merely pre-Hilbert space. Indeed, if $\varphi_u = \varphi_v$, then for all $x \in \mathcal{H}$ one has $0 = \varphi_u(x) - \varphi_v(x) = \langle u - v, x\rangle$, and hence $u = v$ by choosing $x = u - v$ and invoking positive-definiteness of the inner product.

The Riesz theorem then asserts that this map is also surjective. This is non-trivial, and in fact only holds in Hilbert spaces. For example, in an incomplete pre-Hilbert space, surjectivity fails: there exist continuous linear forms that cannot be written as an inner product with any vector in the incomplete space.

The Riesz theorem thus provides the bijection Φ between $\mathcal{H}$ and $\mathcal{H}^*$ that we were seeking. The following corollary is an immediate consequence:

Corollary 2.1

The map $\Phi : u \mapsto \varphi_u$ is an antilinear isometric isomorphism, called the *canonical Riesz isomorphism*, between $\mathcal{H}$ and its topological dual equipped with the dual norm: $\mathcal{H}^* \simeq \mathcal{H}$.

Proof.

1. Antilinearity follows from the antilinearity of the inner product of $\mathcal{H}$: $\Phi(\lambda u) = \varphi_{\lambda u} = \langle \lambda u, .\rangle = \lambda^* \langle u, .\rangle = \lambda^* \varphi_u = \lambda^* \Phi(u)$.
2. We establish the isometry, i.e., $\|\Phi(u)\|_{\mathcal{H}^*} = \|u\|_{\mathcal{H}}$. First:

$$\|\varphi_u\|_{\mathcal{H}^*} = \sup_{\|x\|=1} |\langle u, x\rangle| \leq \|u\|_{\mathcal{H}}$$

by the Cauchy–Schwarz inequality. Equality is attained at $x = u/\|u\|$ when $u \neq 0$, and the case $u = 0$ is trivial.

3. We use the map Φ, a bijective isometry, to transport the inner product of $\mathcal{H}$ to $\mathcal{H}^*$ by defining:
$$\langle \varphi_u, \varphi_v \rangle_{\mathcal{H}^*} \stackrel{\text{def}}{=} \langle \Phi^{-1}(\varphi_v), \Phi^{-1}(\varphi_u) \rangle_{\mathcal{H}} = \langle v, u \rangle_{\mathcal{H}}.$$
Note the reversed order (v, u instead of u, v) to compensate for the antilinearity of Φ. Let us verify sesquilinearity of this inner product on $\mathcal{H}^*$: $\langle \lambda\varphi_u, \varphi_v \rangle_{\mathcal{H}^*} = \langle \varphi_{\lambda^* u}, \varphi_v \rangle_{\mathcal{H}^*} = \langle v, \lambda^* u \rangle_{\mathcal{H}} = \lambda^* \langle v, u \rangle_{\mathcal{H}} = \lambda^* \langle \varphi_u, \varphi_v \rangle_{\mathcal{H}^*}$. The remaining inner product axioms follow easily.

4. The norm induced by this inner product satisfies:
$$\sqrt{\langle \varphi_u, \varphi_u \rangle_{\mathcal{H}^*}} = \sqrt{\langle u, u \rangle_{\mathcal{H}}} = \|u\|_{\mathcal{H}} = \|\varphi_u\|_{\mathcal{H}^*},$$
and thus coincides with the dual norm.

5. Finally, completeness of $\mathcal{H}^*$, and hence its Hilbert space character, is guaranteed by the fact that a bijective isometry from a complete space preserves completeness, which we admit here.

Thus $\mathcal{H}^*$ inherits a Hilbert space structure with respect to which Φ becomes an (anti-linear) isomorphism of Hilbert spaces. □

2.4 Dirac Notation: Bras, Kets, and Brackets

Dirac notation is nothing more than a rewriting of the Riesz isomorphism. We first define kets, bras, and the bracket, then use them to rewrite the Hilbert basis decomposition introduced in the previous lesson.

1. **Kets.** A vector $u \in \mathcal{H}$ is written as a **ket**, denoted:
$$\boxed{u \in \mathcal{H} \quad \longleftrightarrow \quad |u\rangle}$$
The linear structure yields the following computational rules:
$$|u + v\rangle = |u\rangle + |v\rangle \tag{2.1}$$
$$|\lambda u\rangle = \lambda\, |u\rangle \tag{2.2}$$

2. **Bras.** To every vector $u \in \mathcal{H}$ one associates via Φ a form φ_u. This linear form is written as a **bra**, where the associated vector is placed inside the bra:
$$\boxed{\varphi_u = \Phi(u) \in \mathcal{H}^* \quad \longleftrightarrow \quad \langle u|}$$
Thus $\langle u|$ represents the vector u viewed as a linear form, i.e., the operation "take the inner product with u and return a number." The computational rules are:
$$\langle u + v| = \langle u| + \langle v| \tag{2.3}$$
$$\langle \lambda u| = \lambda^*\, \langle u|\,, \tag{2.4}$$
which follow from the antilinearity of Φ.

3. **The bracket.** The two notational conventions above allow one to form an inner product, called a *bracket* (or inner product), as a product of a bra and a ket:

$$\boxed{\langle u, x\rangle \overset{\text{def}}{=} \langle u|x\rangle}$$

This notation reflects the action of the linear form $\langle u|$ on the vector $|x\rangle$:

$$\langle u| \left(|x\rangle\right) = \varphi_u(x) = \langle u, x\rangle = \langle u|x\rangle\,.$$

4. **Hilbert decomposition.** For any Hilbert space dimension, we have seen the decomposition formula $u = \sum_{i\in I}\langle e_i, u\rangle e_i$, where the sum is finite or converges in $\mathcal{H}$ when I is infinite. In Dirac notation this reads:

$$\boxed{|u\rangle = \sum_{i\in I} \langle e_i|u\rangle\,|e_i\rangle = \sum_{i\in I} u_i\,|e_i\rangle} \tag{2.5}$$

where the u_i are the components of the ket u in the basis. Note that they are obtained by orthogonal projection onto e_i as $u_i = \langle e_i|u\rangle$, and *not* as $\langle u|e_i\rangle$, which equals u_i^*. This is a common mistake, likely stemming from the fact that in ordinary vector calculus over $\mathbb{R}^n$ one recovers the component v_i of a vector $\vec{v}$ via $v_i = \vec{v}\cdot\vec{e}_i$, which might suggest $u_i = \langle u|e_i\rangle$ in the quantum case. But this would amount to ignoring the sesquilinearity of the inner product over $\mathbb{C}$, and would immediately lead to a computational error.

For bras one has:

$$\boxed{\langle u| = \sum_{i\in I} \langle u|e_i\rangle\,\langle e_i| = \sum_{i\in I} u_i^*\,\langle e_i|} \tag{2.6}$$

and the squared norm reads:

$$\|u\|^2 = \langle u|u\rangle = \sum_{i\in I} |u_i|^2 = \sum_{i\in I} u_i^* u_i \tag{2.7}$$

where the second equality is Parseval's identity.

5. **The dagger operation.** Applying the isomorphism Φ (from kets to bras) is conventionally denoted by a superscript †. We also write the inverse isomorphism Φ^{-1} from bras to kets using the same symbol, making the dagger operation involutive. By convention:

$$\langle u| = |u\rangle^\dagger \quad \text{(application of } \Phi) \tag{2.8}$$

$$|u\rangle = \langle u|^\dagger \quad \text{(application of } \Phi^{-1}) \tag{2.9}$$

$$\left(|u\rangle^\dagger\right)^\dagger = |u\rangle \quad \text{(involution)} \tag{2.10}$$

Let us illustrate these points in finite dimensions. Take $\mathcal{H}$ to be n-dimensional, with a Hilbert basis $\mathcal{B} = (|e_i\rangle)_{i=1}^n$ equipped with its canonical inner product (cf. Lesson 1, the Hilbert space $\mathbb{C}^n$). Representing the basis kets canonically as column vectors, i.e.,

$(n, 1)$ matrices:

$$|e_i\rangle = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{\mathcal{B}}$$

where the 1 appears in the i-th position, all kets $|u\rangle$ are written as column vectors:

$$|u\rangle = \sum_{i=1}^{n} u_i \, |e_i\rangle = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}_{\mathcal{B}} \in \mathbb{C}^n,$$

with $u_i = \langle e_i | u \rangle$. The linear form φ_u associated to u must satisfy, for every vector v:

$$\varphi_u(v) = \langle u, v \rangle = \sum_{i=1}^{n} u_i^* v_i, \qquad \text{(canonical inner product of } \mathbb{C}^n\text{)}$$

To produce this sum, $\langle u|$ must be represented by the row vector of size $(1, n)$ of conjugated coordinates of u:

$$\langle u| = (u_1^*, u_2^*, \cdots, u_n^*)\,,$$

since the bracket $\langle u|v\rangle$ is then given by the usual matrix product:

$$\langle u|v\rangle = \begin{pmatrix} u_1^*, & u_2^*, & \cdots, & u_n^* \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \sum_{i=1}^{n} u_i^* v_i \in \mathbb{C}.$$

For example, in dimension 2:

$$|u\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \Rightarrow \quad \langle u| = (1, -i)$$

We have obtained the following result:

Proposition 2.1

In finite dimensions, the bra $\langle u|$ is the *conjugate transpose* (also called the Hermitian transpose) of the ket $|u\rangle$:

$$\langle u| = |u\rangle^\dagger = |u^*\rangle^\top \tag{2.11}$$

$$|u\rangle = \langle u|^\dagger = \langle u^*|^\top \tag{2.12}$$

This has no meaning in infinite dimensions, since the transpose is not defined there. That said, in the space $\ell^2(\mathbb{N})$ with its canonical basis, everything works in a broadly similar fashion by considering infinite column or row vectors and replacing finite sums with convergent series. This can be a useful aid to intuition, but strictly speaking these are neither matrices nor transpositions.

2.5 Operators, Matrix Elements, and the Resolution of the Identity

We continue the exposition of Dirac notation, now introducing linear maps (also called linear operators) $\hat{A}$ from $\mathcal{H}$ to $\mathcal{G}$, where $\mathcal{G}$ is another Hilbert space.

Writing operators. First note that in physics it is standard to denote them with a hat (this convention is not used in mathematical texts). An operator $\hat{A}$ acting on a vector v produces the vector $\hat{A}v$. In Dirac notation there are two equivalent ways to write this, the second being the most common:

$$\boxed{v = \hat{A}\,u \quad \longleftrightarrow \quad |v\rangle = |\hat{A}u\rangle = \hat{A}\,|u\rangle}.$$

Likewise, an inner product involving an operator can be written as:

$$\boxed{\langle w, \hat{A}\,u\rangle \quad \longleftrightarrow \quad \langle w|\hat{A}u\rangle = \langle w|\,\hat{A}\,|u\rangle}$$

where the second form is more widely used for aesthetic reasons.

Matrix elements. In the formula above, the case $\langle w| = \langle e_i|$ and $|u\rangle = |e_j\rangle$ is particularly important, as it defines the *matrix elements* of the operator $\hat{A}$:

$$\boxed{A_{ij} = \langle e_i|\,\hat{A}\,|e_j\rangle\,.} \tag{2.13}$$

In infinite dimensions, this expression is not always well-defined, since one must also require that $|e_i\rangle$ belongs to the domain of $\hat{A}$; we will return to this point later. In finite dimensions, however, the definition is transparent: via the column-vector representation of basis vectors, a linear operator is in one-to-one correspondence with its representing matrix; we therefore write $\hat{A} \longleftrightarrow A = (A_{ij})_{1\leq i,j\leq n}$ with $A_{ij} = \langle e_i|\,\hat{A}\,|e_j\rangle$. The equation $|v\rangle = \hat{A}\,|u\rangle$ then corresponds to the usual matrix product $\hat{A}\,|u\rangle = \sum_i \sum_j (A_{ij}u_j)\,|e_i\rangle$, while an inner product is written as $\langle w|\,\hat{A}\,|u\rangle = (w^*)^\top Au = \sum_{ij} w_i^* A_{ij} u_j \in \mathbb{C}$.

Example 2.1

Explicitly, suppose $\quad \hat{A} = \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix}, \quad |u\rangle = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad |w\rangle = \begin{pmatrix} i \\ 1 \end{pmatrix},$

then:

$$|v\rangle = \hat{A}\,|u\rangle = \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} i \\ -2 \end{pmatrix}, \quad \langle w|\,\hat{A}\,|u\rangle = (-i, 1) \cdot \begin{pmatrix} i & 0 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} = -1.$$

Ket-bra operators. A bracket is a number, but a ket-bra is an operator, called an outer product (unrelated to the exterior product of differential forms). Given two vectors $|u\rangle$ and $|v\rangle$ in the *same* Hilbert space $\mathcal{H}$, one defines the operator $|u\rangle\langle v|$ from $\mathcal{H}$ to $\mathcal{H}$ by:

$$|u\rangle\langle v| \quad : \quad \mathcal{H} \to \mathcal{H}, \tag{2.14}$$

$$|x\rangle \mapsto \underbrace{\langle v|x\rangle}_{\in\mathbb{C}}\,|u\rangle\,. \tag{2.15}$$

Again, *in finite dimensions*, the case $\hat{E}_{ij} \stackrel{\text{def}}{=} |e_i\rangle\langle e_j|$ is particularly important. This is the operator whose matrix elements are all zero except for a single entry of 1 in row i and column j; its associated matrix is:

$$E_{ij} = \begin{pmatrix} 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{pmatrix},$$

where the 1 is located at row i and column j. Every linear operator $\hat{A}$ then decomposes according to its matrix elements:

$$\boxed{\hat{A} = \sum_{i,j=1}^{n} A_{ij}\, |e_i\rangle\langle e_j|\,,}$$

where $A_{ij} = \langle e_i|\,\hat{A}\,|e_j\rangle$. In infinite dimensions, this decomposition does not always exist; when it does, it must be understood in the sense of strong convergence (at minimum). We will return to this in subsequent lessons.

Decomposition of the identity operator. The identity operator on $\mathcal{H}$ is denoted $\mathbb{1}$ (or simply I), and is trivially defined by $\mathbb{1}\,|\psi\rangle = |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}$. Its matrix elements in any orthonormal basis are δ_{ij} (the Kronecker delta). In finite or countably infinite dimensions, one has the formula

$$\boxed{\mathbb{1} = \sum_{i\in I} |e_i\rangle\langle e_i|} \tag{2.16}$$

which converges in the strong sense. It is called the *resolution of the identity* or *closure relation*. This formula is fundamental to all computations in quantum mechanics.

2.6 The Adjoint Operator

We return to the Riesz isomorphism and consider a continuous linear operator[2] $\hat{A}$ from $\mathcal{H}$ to $\mathcal{G}$, where $\mathcal{H}$ and $\mathcal{G}$ are two Hilbert spaces, possibly equal.

When $\hat{A}$ acts on a ket $|u\rangle \in \mathcal{H}$, it produces a new ket $|v\rangle = |\hat{A}u\rangle = \hat{A}\,|u\rangle \in \mathcal{G}$. It is natural to ask what the Riesz-associated bra of $|v\rangle$ is, i.e., what the linear form on $\mathcal{G}$ given by $\langle v| = \langle \hat{A}u|$ equals. This question leads naturally to a new operator, called the *adjoint of* $\hat{A}$, again denoted $\hat{A}^\dagger$. This operator plays an essential role in quantum mechanics.

To determine $\langle v| = \langle \hat{A}u|$, fix an arbitrary ket $|w\rangle \in \mathcal{G}$ and consider the linear form on $\mathcal{H}$:

$$\varphi : \begin{array}{ccc} \mathcal{H} & \longrightarrow & \mathbb{C} \\ |u\rangle & \longmapsto & \langle \hat{A}u, w\rangle_{\mathcal{G}} \end{array}$$

[2]We restrict to continuous operators here for simplicity. The construction of the adjoint in the non-continuous case will be addressed later (cf. the subsequent lessons on linear operators and on unbounded operators).

(Here the subscript $\mathcal{G}$ indicates in which space the inner product is taken.) We admit that φ is a continuous linear form on $\mathcal{H}$ whenever $\hat{A}$ is a continuous operator. By the Riesz theorem applied to $\mathcal{H}$, there exists a unique vector $|z\rangle \in \mathcal{H}$ such that $\varphi = \varphi_z$, i.e.:

$$\forall\ |u\rangle \in \mathcal{H}, \quad \varphi(u) = \langle \hat{A}u, w\rangle_{\mathcal{G}} = \varphi_z(u) = \langle u, z\rangle_{\mathcal{H}}.$$

The above construction thus associates to each vector $|w\rangle \in \mathcal{G}$ a unique vector $|z\rangle \in \mathcal{H}$. This operation can be formally expressed as the action of an operator—the *adjoint operator*, denoted $\hat{A}^\dagger$—via $|z\rangle = \hat{A}^\dagger\,|w\rangle$. Note that whereas $\hat{A}$ maps $\mathcal{H}$ to $\mathcal{G}$, the adjoint maps $\mathcal{G}$ to $\mathcal{H}$.

Carrying out this construction for all $|w\rangle$, one readily verifies that the operator so defined is indeed a continuous linear operator. This yields the following fundamental result:

Definition 2.2 : Adjoint of a continuous operator

The adjoint of a continuous linear operator $\hat{A} : \mathcal{H} \to \mathcal{G}$ is the unique continuous linear operator $\hat{A}^\dagger : \mathcal{G} \to \mathcal{H}$ such that:

$$\boxed{\forall\ |u\rangle \in \mathcal{H},\ \forall\ |w\rangle \in \mathcal{G}, \quad \langle \hat{A}u|w\rangle_{\mathcal{G}} = \left\langle u\middle|\hat{A}^\dagger w\right\rangle_{\mathcal{H}}} \tag{2.17}$$

The adjoint has two practical uses in formal quantum computations.

1. The formula above shows that, loosely speaking, "with u and w held fixed, the adjoint allows one to move the operator from the left to the right argument of the inner product."
2. Using the Hermitian symmetry of the inner product, $\langle \hat{A}u|w\rangle_{\mathcal{G}} = \langle w|\hat{A}u\rangle_{\mathcal{G}}^*$, one also obtains:
$$\boxed{\forall\ |u\rangle \in \mathcal{H},\ \forall\ |w\rangle \in \mathcal{G}, \quad \langle w|\,\hat{A}\,|u\rangle_{\mathcal{G}}^* = \langle u|\,\hat{A}^\dagger\,|w\rangle_{\mathcal{H}}} \tag{2.18}$$
which shows that "the adjoint allows one to exchange bra and ket, up to complex conjugation."

These two identities are equivalent. One must of course keep track of the spaces $\mathcal{H}$ and $\mathcal{G}$, and of which inner product—that of $\mathcal{H}$ or that of $\mathcal{G}$—is being used. In practice, however, one almost always has $\mathcal{H} = \mathcal{G}$, and the notation simplifies accordingly.

We now illustrate the preceding construction in finite dimensions. The main result is the following:

Proposition 2.2

In finite dimensions, the matrix representing the adjoint $\hat{A}^\dagger$ is the conjugate transpose of the matrix representing the operator $\hat{A}$:

$$\boxed{A^\dagger = (A^*)^\top} \tag{2.19}$$

Proof. The matrix elements of $\hat{A}^\dagger$ are by definition:

$$(\hat{A}^\dagger)_{ij} = \langle e_i|\,\hat{A}^\dagger\,|e_j\rangle = \left\langle e_i\middle|\hat{A}^\dagger e_j\right\rangle.$$

By definition of the adjoint:

$$\left\langle e_i \middle| \hat{A}^\dagger e_j \right\rangle = \langle \hat{A} e_i | e_j \rangle$$

and by Hermitian symmetry:

$$\langle \hat{A} e_i | e_j \rangle = \langle e_j | \hat{A} e_i \rangle^* = \langle e_j | \hat{A} | e_i \rangle^* = A^*_{ji}$$

We have therefore shown:

$$(A^\dagger)_{ij} = A^*_{ji} = \left(A^{*\top} \right)_{ij}$$

$\square$

We have not yet answered the question explicitly: what is the bra $\langle v| = \langle \hat{A}u|$ canonically associated with the ket $|v\rangle = |\hat{A}u\rangle$? Equation (2.17) answers this via an inner product valid for all w:

$$\forall w \in \mathcal{G}, \quad \left\langle \hat{A}u \middle| w \right\rangle_{\mathcal{G}} = \langle u| \hat{A}^\dagger |w\rangle_{\mathcal{H}} \Rightarrow \boxed{\left\langle \hat{A}u \right| = \langle u| \hat{A}^\dagger}$$

Warning: a common confusion is to think that "the adjoint acts on the left on bras." This is incorrect. The expression $\langle u| \hat{A}^\dagger$ is a *composition of operators*, not a left action. The relevant spaces are:

$$\hat{A}^\dagger : \mathcal{G} \to \mathcal{H},$$
$$\langle u| : \mathcal{H} \to \mathbb{C},$$

so that $\langle u| \hat{A}^\dagger$ is the composition $\langle u| \circ \hat{A}^\dagger : \mathcal{G} \to \mathcal{H} \to \mathbb{C}$, which is indeed a linear form on $\mathcal{G}$. In general the symbol $\circ$ is omitted, which can be a source of confusion.

Example 2.2 : Illustration in finite dimensions

Kets are column vectors, bras are row vectors, and the adjoint is the conjugate transpose. Take, for instance:

$$\hat{A} = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix}, \quad \hat{A}^\dagger = \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix}, \quad |u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We compute:

$$\hat{A}|u\rangle = \begin{pmatrix} 1 & i \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Rightarrow \quad \langle \hat{A}u| = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

Let us verify $\langle u| \hat{A}^\dagger$ (the row-times-matrix product):

$$\langle u| \hat{A}^\dagger = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -i & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

We recover the same row vector. Here $\langle u| \hat{A}^\dagger$ is literally a row-times-matrix product, not a left action of $\hat{A}^\dagger$ on $\langle u|$, which would be meaningless as a matrix operation.

Finally, from $\langle \hat{A}u| = \langle u| \hat{A}^\dagger$, we deduce the following useful identities by applying the dagger:

$$\left(\hat{A}\left|u\right\rangle\right)^\dagger = \left\langle u\right| \hat{A}^\dagger, \tag{2.20}$$

$$\left(\left\langle u\right| \hat{A}^\dagger\right)^\dagger = \hat{A}\left|u\right\rangle. \tag{2.21}$$

Example 2.3

Consider a typical elementary quantum mechanics exam question on the harmonic oscillator. The problem provides a family of kets $|n\rangle$ for non-negative integers n, together with the relations $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$ and $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$. The question asks: what is $\langle n|\,\hat{a}^\dagger$?

If one mistakenly believes that $\hat{a}^\dagger$ acts to the left, one might be tempted to answer $\langle n|\,\hat{a}^\dagger = \sqrt{n+1}\,\langle n+1|$, which is wrong. Using the identities derived above:

$$\langle n|\,\hat{a}^\dagger = (\hat{a}\,|n\rangle)^\dagger = \left(\sqrt{n}\,|n-1\rangle\right)^\dagger = \sqrt{n}\,\langle n-1|$$

The following figure summarises the relationships between bras, kets, the operator A, and its adjoint, as developed in this chapter.

We close this section with a few important properties of the adjoint operation:

Proposition 2.3 : Properties of the adjoint operator.

For all continuous linear operators $\hat{A}, \hat{B}$ and all scalars $\lambda \in \mathbb{C}$:

$$(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger, \tag{2.22}$$

$$(\lambda \hat{A})^\dagger = \lambda^* \hat{A}^\dagger, \qquad \text{(antilinearity)} \tag{2.23}$$

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger, \qquad \text{(note the reversed order)} \tag{2.24}$$

$$(\hat{A}^\dagger)^\dagger = \hat{A}, \qquad \text{(involution).} \tag{2.25}$$

Proof. (Composition case): for all $|u\rangle \in \mathcal{H}$ and $|w\rangle \in \mathcal{G}$, we apply the fundamental identity Eq. (2.17) twice:

$$\left\langle (\hat{A}\hat{B})u \middle| w \right\rangle = \left\langle \hat{A}(\hat{B}u) \middle| w \right\rangle = \left\langle \hat{B}u \middle| \hat{A}^\dagger w \right\rangle = \left\langle u \middle| \hat{B}^\dagger (\hat{A}^\dagger w) \right\rangle = \left\langle u \middle| (\hat{B}^\dagger \hat{A}^\dagger) w \right\rangle .$$

We conclude by uniqueness of the adjoint. $\square$

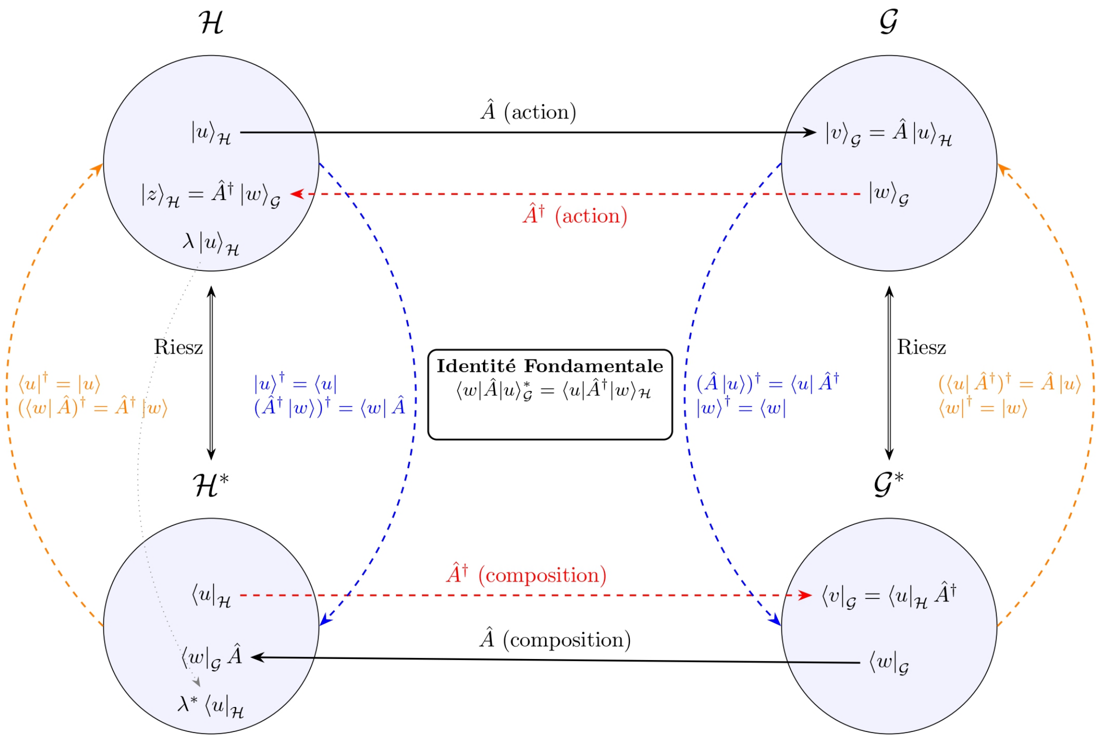

Figure 2.1: Representation of the source Hilbert space $\mathcal{H}$ and target Hilbert space $\mathcal{G}$, together with their topological duals $\mathcal{H}^*$ and $\mathcal{G}^*$. The operator $\hat{A}$ has a natural right action: it maps a ket in the source space $|u\rangle_{\mathcal{H}}$ to a ket in the target space $\hat{A}\,|u\rangle_{\mathcal{H}} \in \mathcal{G}$. The Riesz isomorphism allows one to construct its adjoint $\hat{A}^\dagger$, which acts naturally on kets from $\mathcal{G}$ to $\mathcal{H}$ and satisfies the *fundamental identity*. Warning: the diagram should not suggest that this is the inverse map; in general $\hat{A}^\dagger \neq \hat{A}^{-1}$. The upper horizontal arrows represent actions of operators, while the lower horizontal arrows represent compositions as discussed in the text. The blue and orange arrows indicate the dagger operation (the Riesz isomorphism Φ or its inverse Φ^{-1}). The relevant formulas for these operations are given in the text (where the subscripts $\mathcal{H}$ and $\mathcal{G}$ have been omitted for readability). Since this operation is involutive, the blue and orange transformations are inverses of each other. The grey dashed line is a reminder of the antilinearity of this isomorphism, which is the source of the complex conjugation in the fundamental identity. The subscripts $\mathcal{H}$ and $\mathcal{G}$ in this identity specify in which space each inner product is computed.